

ESTIMATION OF THE NUMBER OF CLASSES

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Summary: A population is partitioned into S classes, and a sample of N individuals is drawn. The number of classes in which no individuals appear is unobservable, and the estimation of this quantity, or alternatively the total number of classes S , has been considered by many authors. Sichel (1986a, 1986b) considered the problem in the context of word frequency distributions and the estimation of an author's vocabulary. Using a mixed Poisson model for species abundance, Sichel (1992) proposed an estimator for the total number of species, which was also arrived at by Ord and Whitmore (1986). In this paper we show that these estimators are equivalent under maximum likelihood estimation, give a method for maximum likelihood estimation, derive the asymptotic variances of the parameter estimators and $\hat{S}$, and investigate by simulation the sampling distribution of $\hat{S}$.

1. Introduction

The estimation of the number of classes present in a population, where individuals are partitioned into one of a finite number of classes, has received attention in the literature in a number of contexts, the most well-known being the number of species (Ord & Whitmore, 1986, Sichel, 1992) and the size of an author's vocabulary (Efron & Thisted, 1976, Sichel, 1986a, 1986b). The topic was reviewed by Bunge and Fitzpatrick (1993).

Sichel (1992) proposed an estimator for the total number of species, S , based on a mixed Poisson model (Generalized Inverse Gaussian Poisson or GIGP) for the distribution of individuals within species; this estimator was arrived at independently by Ord and Whitmore (1986) for the Poisson-Inverse Gaussian (P-IG) special case. Sichel (1986a, 1986b) also developed a model for word frequency distributions using occupancy theory, and proposed an estimator ($\hat{V}$) for an author's total vocabulary.

In this paper we restate the estimators for the total number of classes in terms of an advantageous parameterization of the GIGP introduced by Stein, Zucchini and Juritz (1987); prove that the maximum likelihood estimators $\hat{S}$ and $\hat{V}$ are equal; give a technique for maximum likelihood estimation for the zero-truncated GIGP; derive the asymptotic variance of the estimator; and evaluate its small-sample behaviour.

2. The models

The population is partitioned into S classes. We have available a random sample of N individuals, with X_i coming from the i th class, $i=1, \dots, n_N$. Essentially our purpose is to estimate the number of classes in which no individuals were observed, or alternatively the total number of classes S . We now outline two models which appear in the literature, one for species frequency distribution, the other for word frequency distribution. While these models have different origins, the resultant distributions for the number of individuals in a class are the same.

2.1 Species frequency distribution

The compound Poisson model is described in detail, in terms of an inverse Gaussian mixing distribution, by Ord and Whitmore (1986). In terms of a generalized inverse Gaussian mixing distribution:

- The number of individuals X_i observed in class i is distributed as Poisson ($\lambda = \lambda_i$) $i = 1, \dots, S$
- The λ_i are drawn independently from a generalized inverse Gaussian distribution, which is (using Stein *et al.*, 1987 parameterization):

$$f(\lambda) = \frac{\xi^{-\gamma}}{2K_\gamma(\omega)} \lambda^{\gamma-1} \exp\left\{-\frac{\omega}{2}\left(\frac{\lambda}{\xi} + \frac{\xi}{\lambda}\right)\right\} \quad \lambda > 0$$

(The inverse Gaussian distribution is a special case of the above with $\gamma = -\frac{1}{2}$.)

We denote this distribution as $GIG(\omega, \xi, \gamma)$.

The resulting Poisson mixture is the generalized inverse Gaussian Poisson (GIGP) (Sichel, 1971, Stein *et al.*, 1987), which has also been called the Sichel distribution by Johnson *et al.* (1992) and Stein *et al.* (1987):

$$P(X_i = x) = \frac{\left(\frac{\omega}{\alpha}\right)^\gamma}{K_\gamma(\omega)} \frac{\left(\frac{\xi\omega}{\alpha}\right)^x}{x!} K_{x+\gamma}(\alpha) \quad x = 0, 1, 2, \dots$$

where $\alpha = \sqrt{\omega(\omega + 2\xi)}$ and $K_\nu(z)$ is a modified Bessel function of the third kind (Abramowitz and Stegun 1972). Stein *et al.* (1987) show that, for the case $\gamma = -\frac{1}{2}$, ξ is the population mean and α is a shape parameter.

Introducing the notation

$$g_\nu(z) = z^{-\nu} K_\nu(z)$$

the probability function simplifies to

$$P(X_i = x) = \frac{g_{x+\gamma}(\alpha)}{g_\gamma(\omega)} \frac{(\xi\omega)^x}{x!} \quad x = 0, 1, 2, \dots \quad (1)$$

We denote this as GIGP (ω, ξ, γ) , having

$$E(X_i) = \xi \frac{K_{\gamma+1}(\omega)}{K_{\gamma}(\omega)}$$

and

$$\text{Var}(X_i) = E(X_i) \left\{ 1 + \xi \left[\frac{K_{\gamma+2}(\omega)}{K_{\gamma+1}(\omega)} - \frac{K_{\gamma+1}(\omega)}{K_{\gamma}(\omega)} \right] \right\}$$

The GIGP appears in the literature in a number of parameterizations (Sichel, 1971, Atkinson & Lam Yeh, 1982, Stein *et al.*, 1987); the one we have used above has the advantage that the maximum likelihood estimators $\hat{\alpha}$ and $\hat{\xi}$ are asymptotically uncorrelated in the case $\gamma = -\frac{1}{2}$, and only mildly correlated for a reasonable range of γ (Stein *et al.*, 1987). However, since the zero frequencies are unobservable, we are dealing with the zero-truncated GIGP:

$$P^*(X_i = x) = \frac{1}{1 - p_0} P(X_i = x) \quad x = 1, 2, \dots \quad (2)$$

$$\begin{aligned} \text{where } p_0 &= P(X_i = 0) \\ &= g_{\gamma}(\alpha)/g_{\gamma}(\omega). \end{aligned}$$

The above model implies that sampling is being performed over a unit time period. In order to make explicit the dependence of the distribution of X_i on the sampling period, we note that X_i observed over a time period t is distributed as Poisson($\lambda_i t$), where λ_i is the mean occurrence rate in a unit time period. Sichel (1992) showed that, as the number of individuals N in the sample (representing "sampling effort" or "text length") is proportional to t in the long run, t may be replaced by N ; the parameter ω (b in his notation) is invariant to N , as is γ ; and that the distribution of the number of individuals in the i th class, given a sample of N individuals, is GIGP $(\omega, N\xi, \gamma)$. In this case, the appropriate α parameter is $\alpha_N = \sqrt{\omega(\omega + 2N\xi)}$.

- Ord and Whitmore (1986) present the maximum likelihood estimator

$$\hat{S} = \frac{n_N}{1 - \hat{p}_0} \quad (3)$$

where $\hat{p}_0$ is the estimated proportion of classes that were not represented in the sample, based on the assumption of an inverse Gaussian mixing distribution. Sichel (1992) arrives at the same estimator using the GIG.

2.2 Word frequency distribution

Sichel (1986a) developed the following model for word frequency distributions:

- An author's total vocabulary is V .
- Each word has a long-term probability π_i of being used, and these probabilities are sorted as $\pi_1, \pi_2, \dots, \pi_V$ in ascending order. (For example, in English "the" is the most commonly-used word, with $\pi_V \simeq 0.07$). Typically V is of the order of several thousand, so the π 's are crowded in the small interval $0 \leq \pi_i \leq 0.07$.
- The frequency of the π 's is approximated by a continuous density, in this case the generalized inverse Gaussian distribution $\text{GIG}(\omega, \xi, \gamma)$. Using a binomial distribution for the number of times a word appears in a text of length N , Sichel (1986a) shows that V is the inverse of the mean of the distribution of the π_i 's. He obtains parameter estimates by equating the sample mean and proportion of singletons to their population equivalents; for the generalized inverse Gaussian, this gives the estimator

$$\hat{V} = [\hat{\xi} K_{\hat{\gamma}+1}(\hat{\omega}) / N K_{\hat{\gamma}}(\hat{\omega})]^{-1} \quad (4)$$

Using a Poisson approximation for the binomial distribution gives a resulting GIGP $(\omega, N\xi, \gamma)$ distribution for the number of words of the i -th word type in a text of length N words.

3. Maximum likelihood estimation

Under the models described above, it is assumed that the number of individuals N is fixed, and the observed number of classes, n_N , is random. n_N corresponds to the sample size n , which is fixed in classical applications, and N to $\sum_{i=1}^{n_N} x_i$, which is conventionally random.

Using the notation $f_n(\underline{x})$ for the density function of $\underline{x}$ with n fixed and known, the joint density of the sample $\underline{x}$, under fixed n_N , is the usual

$$f_{n_N}(\underline{x}|\alpha, \xi, \gamma) = \prod_{i=1}^{n_N} f(x_i|\alpha, \xi, \gamma).$$

Once n_N is random we have the joint density of $\underline{x}$ and n_N

$$f_N(\underline{x}, n_N|\alpha, \xi, \gamma) = f_{n_N}(\underline{x}|\alpha, \xi, \gamma) \cdot g(n_N|\alpha, \xi, \gamma, N) \quad (5)$$

where $g(\cdot)$ is the probability function of n_N . The likelihood corresponding to (5) is

$$l_N(\alpha, \xi, \gamma|\underline{x}, n_N) = \sum_{i=1}^{n_N} \ln f(x_i|\alpha, \xi, \gamma) + \ln g(n_N|\alpha, \xi, \gamma, N). \quad (6)$$

We show in the Appendix that the distribution of n_N given N tends to normality with

$$\mu_{n_N} = [\xi R_\gamma(\omega)]^{-1}$$

and

$$\sigma_{n_N}^2 = \frac{1 + N\xi[R_{\gamma+1}(\omega) - R_\gamma(\omega)]}{N\{\xi R_\gamma(\omega)\}^2}$$

This gives the second term on the right-hand side of (6).

Usual maximum likelihood estimation maximizes the first term on the right-hand side of (6): for example Ord and Whitmore's (1986) $\hat{S}$. In this paper we

will be doing the same, i.e. omitting the term $\ln . g(n_N|\alpha, \xi, \gamma, N)$ from the likelihood. We assume that this term is negligible; we will see the consequences of this omission when we simulate the sampling distribution of $\hat{S}$.

The Bessel functions $K_\nu(z)$ render m.l. estimation a nontrivial computational exercise; in particular $\frac{\partial K_\nu(z)}{\partial \nu}$ is not available in closed form, and numerical approximation of this derivative may not be accurate as $K_\nu(z)$ grows extremely large with increasing ν . Stein *et al.* (1987) give a procedure for m.l. estimation of α, ξ and γ for the (untruncated) GIGP, based on Newton-Raphson iteration for $\hat{\alpha}$ and $\hat{\xi}$ embedded in a line search for $\hat{\gamma}$; The need for derivatives with respect to the order in the Bessel functions is thus obviated. Burrell and Fenton (1993) employ a numerical method for m.l. estimation in the zero-truncated case.

We used Stein *et al.*'s (1987) method, extended to the zero-truncated GIGP. The optimization was performed in S-Plus (Becker *et al.*, 1988) using the *nlimb* function, which employs a trust region approach to linearly constrained optimization (Gay, 1984). This utilises the first and second derivatives of the log likelihood function with respect to α and ξ ; these are given in the Appendix as (B.1) to (B.5).

Given the m.l.e.'s $\hat{\alpha}, \hat{\xi}$ and $\hat{\gamma}$, we compute

$$\hat{\omega} = \sqrt{\hat{\alpha}^2 + \hat{\xi}^2} - \hat{\xi}$$

and by the invariance property of maximum likelihood estimators, (B.2) is the m.l.e. of S , with $\hat{p}_0 = g_\gamma(\hat{\alpha})/g_\gamma(\hat{\omega})$. Similarly (4) is the m.l.e. of V .

However, the likelihood equation for $\hat{\xi}$ (B.2) can be stated as

$$\frac{\bar{x}}{\hat{\xi}} = \frac{\hat{\omega} g_{\hat{\gamma}+1}(\hat{\omega})}{d_{\hat{\gamma}}(\hat{\alpha}, \hat{\xi})} \quad (7)$$

where

$$d_{\gamma}(\alpha, \xi) = g_{\gamma}(\omega) - g_{\gamma}(\alpha)$$

$$\begin{aligned} \text{Now } \hat{S} &= \frac{n_N}{1 - g_{\hat{\gamma}}(\hat{\alpha})/g_{\hat{\gamma}}(\hat{\omega})} \\ &= \frac{n_N}{d_{\hat{\gamma}}(\hat{\alpha}, \hat{\xi})} \cdot g_{\hat{\gamma}}(\hat{\omega}) \end{aligned}$$

and from (4), and using $N = n_N \bar{x}$,

$$\hat{V} = \frac{n_N \bar{x}}{\hat{\xi}} \frac{K_{\hat{\gamma}}(\hat{\omega})}{K_{\hat{\gamma}+1}(\hat{\omega})}$$

Substituting (7),

$$\begin{aligned} \hat{V} &= n_N \frac{\hat{\omega} g_{\hat{\gamma}+1}(\hat{\omega})}{d_{\hat{\gamma}}(\hat{\alpha}, \hat{\xi})} \frac{K_{\hat{\gamma}}(\hat{\omega})}{K_{\hat{\gamma}+1}(\hat{\omega})} \\ &= \frac{n_N}{d_{\hat{\gamma}}(\hat{\alpha}, \hat{\xi})} \hat{\omega} \hat{\omega}^{-(\hat{\gamma}+1)} K_{\hat{\gamma}+1}(\hat{\omega}) K_{\hat{\gamma}}(\hat{\omega}) / K_{\hat{\gamma}+1}(\hat{\omega}) \\ &= \frac{n_N}{d_{\hat{\gamma}}(\hat{\alpha}, \hat{\xi})} g_{\hat{\gamma}}(\hat{\omega}) \\ &= \hat{S} \end{aligned}$$

and the m.l.e.'s for the two apparently different expressions for the total number of classes are in fact identical.

3.1 Example : Filarial worms on mites on rats

Bertram (1949) fed 2600 mites on rats infected with Cotton Rat Filariasis, examined them microscopically after feeding, and recorded the number of filarial worms on each mite. The zero frequency was 1155, i.e. 1155 mites had no filarial worms. Sichel (1971) fitted the P-IG model to these data; Rubinstein (1985) obtained an improved fit using the GIGP model:

TABLE I. M.L. estimates for Filarial worms data

Estimate	P-IG	GIGP
$\hat{\alpha}$	0.997	0.960
$\hat{\xi}$	2.608	2.410
$\hat{\gamma}$	-	-0.464
Likelihood	-5040.1	-5040.0

In order to gauge how well the estimation proceeds on zero-truncated data, we omitted the zero frequency and obtained the following estimates using the above method:

TABLE II. M.L. estimates for zero-truncated Filarial worms data

Estimate	P-IG	GIGP
$\hat{\alpha}$	1.012	0.896
$\hat{\xi}$	2.635	2.210
$\hat{\gamma}$	-	-0.438
$\hat{p}_0$	0.439	0.459
$\hat{S}$	2575.8	2669.3
Likelihood	-5040.2	-5041.0

The likelihood values shown in Table II are those for the full data set, computed at the parameter values arrived at through estimation on the truncated data. Clearly the likelihood differences are not significant.

Considering that we have omitted $\frac{1155}{2600} = 44\%$ of the sample by truncating the zero frequency, the closeness of these estimates to those using all of the data is pleasing.

4. Asymptotic variance of the estimators

The asymptotic variances and covariance of $\hat{\alpha}$ and $\hat{\xi}$, for fixed γ and n_N , are derived in the appendix as (B.6) to (B.8).

We then have

$$\begin{aligned} asvar(\hat{S}|\gamma, n_N) &= asvar(\hat{\alpha}|\gamma, n_N)\left(\frac{\partial S}{\partial \alpha}\right)^2 + asvar(\hat{\xi}|\gamma, n_N)\left(\frac{\partial S}{\partial \xi}\right)^2 \\ &\quad + 2 ascov(\hat{\alpha}, \hat{\xi}|\gamma, n_N)\left(\frac{\partial S}{\partial \alpha}\right)\left(\frac{\partial S}{\partial \xi}\right) \\ \text{where } \frac{\partial S}{\partial \alpha} &= n_N \frac{[-\alpha g_\gamma(\omega) g_{\gamma+1}(\alpha) + \omega g_\gamma(\alpha) g_{\gamma+1}(\omega) \frac{\alpha}{\sqrt{\xi^2 + \alpha^2}}]}{d_\gamma(\alpha, \xi)^2} \\ \text{and } \frac{\partial S}{\partial \xi} &= \frac{-n_N \omega^2 g_\gamma(\alpha) g_{\gamma+1}(\omega)}{(d_\gamma(\alpha, \xi))^2 \sqrt{\xi^2 + \alpha^2}} \end{aligned}$$

While it is usual to express the variance of estimators in terms of the sample size n_N , because n_N is random we need to express the asymptotic variance in terms of N . We note that this is of the form

$$asvar(\hat{S}|\gamma, n_N) = n_N \cdot f(\alpha, \xi, \gamma)$$

and taking the expectation over n_N ,

$$\begin{aligned} E[asvar(\hat{S}|\gamma)] &= \sum_{n_N=0}^N n_N \cdot f(\alpha, \xi, \gamma) \cdot g(n_N) \\ &= f(\alpha, \xi, \gamma) \cdot E(n_N) \end{aligned} \quad (8)$$

Using an urn model, Sichel (1986b) gives the asymptotic distribution of n_N as normal with

$$E(n_N) \simeq S \left[1 - \int_0^1 e^{-N\pi} \psi(\pi) d\pi \right] \quad (9)$$

where $\psi(\cdot)$ is the density function of π , the probability of a word appearing in the text, or alternatively the probability of an animal of a particular species being sampled.

Sichel (1986b) gives the form of (9) in the case of $\psi(\cdot)$ being the inverse Gaussian density; in the case of the generalized inverse Gaussian, this is

$$\begin{aligned} E(n_N) &\simeq S[1 - g_\gamma(\alpha_N)/g_\gamma(\omega)] \\ &= S(1 - p_0) \end{aligned} \quad (10)$$

which one would expect from (3). (Alternatively, we have from (3) that $E(n_N) \simeq S$; these are asymptotically equivalent.) Substitution of (10) into (8) gives the expected asymptotic variance of $\hat{S}$, for fixed γ .

Example: Filarial worms on mites on rats

We have estimated γ , so $asvar(\hat{S}|\gamma)$ will be an underestimate. Nevertheless we compute

$$asvar(\hat{S}|\gamma = \hat{\gamma}) = 20492$$

giving a 95% confidence interval for S of (2389, 2950).

The asymptotic variances and covariances of the estimators when γ is estimated involve derivatives with respect to the order of the Bessel functions. As was mentioned in Section 3, these derivatives are not available in closed form; nor is it advisable to approximate them numerically.

A simulation study to assess the sampling distribution of $\hat{S}$, for γ both fixed and unknown, over a range of N , was performed.

5. Simulation

A simulation study was conducted to assess the convergence of the sampling distribution of $\hat{S}$ to its asymptotic form, for fixed γ , and to investigate its sampling distribution when γ is estimated.

Samples were generated over a range of (α, ξ, γ) , and a range of fixed n_N : the variance of $\hat{S}$ agreed closely with $asvar(\hat{S}|\gamma, n_N)$ for fixed γ , and was somewhat greater than $asvar(\hat{S}|\gamma = \hat{\gamma}, n_N)$ for γ estimated, as expected.

However, once samples were generated with $\Sigma x_i = N$ fixed, the sampling distribution of $\hat{S}$ was more troublesome. Figure 1 depicts the simulated mean and variance of $\hat{S}$, for 100 simulated samples drawn from the GIGP distribution with $\alpha = 1, \xi = 0.001N, \gamma = -\frac{1}{2}$ and $N=500, 1000, 1500, 2000, 5000$ and 10000 . Parameter estimates were computed assuming (a) γ fixed, and (b) γ unknown. It can be seen that $\hat{S}$ shows some upward bias for large N , and the variance for both γ fixed and γ estimated converge to a limit greater than that given by the asymptotic formula.

When N is very large, $p_0 \simeq 0$ and $\hat{S} \simeq n_N$. The approach of $\hat{S}$ to n_N is depicted, for $N=1500, 2000, 5000$ and 10000 , in Figure 2.

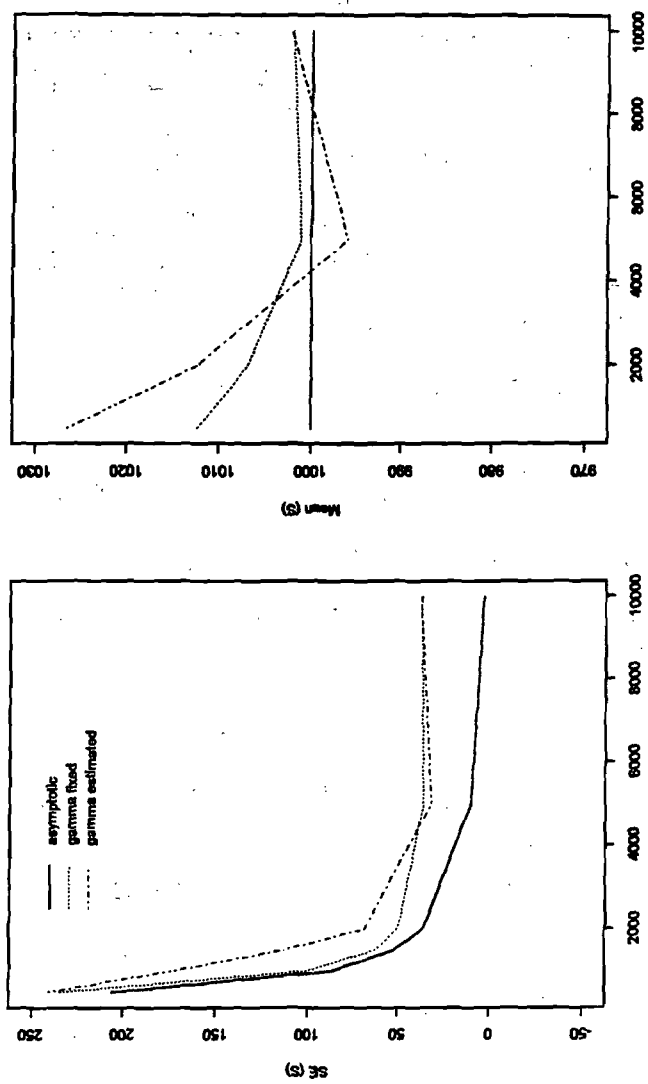
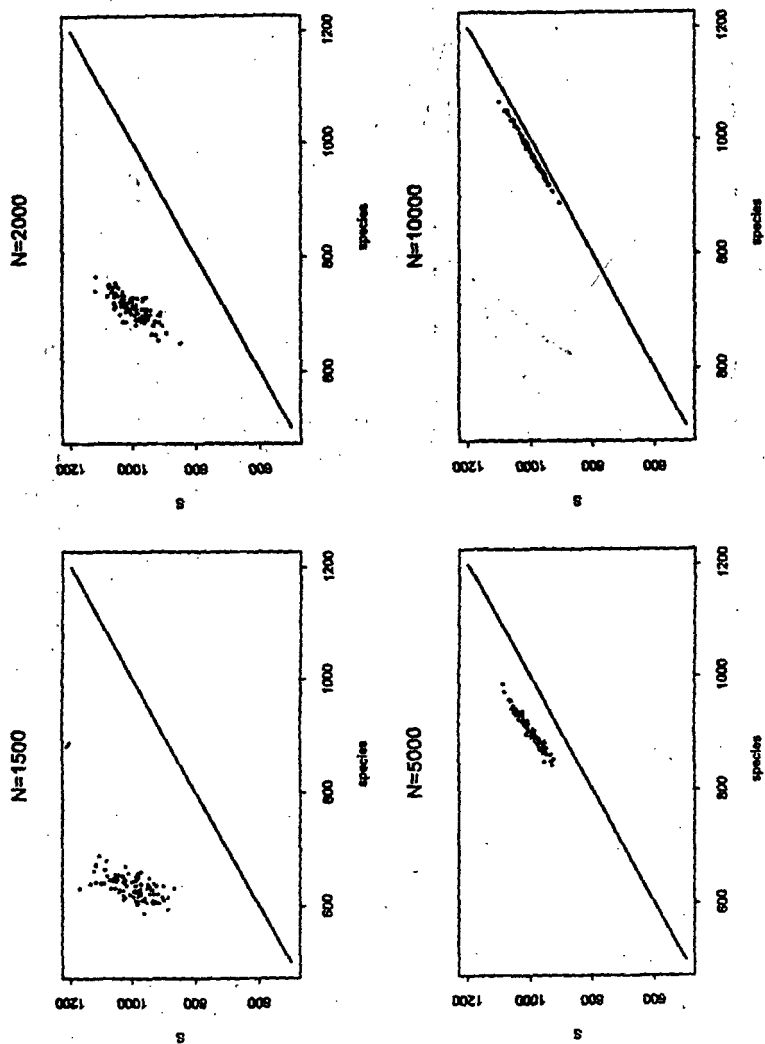


Figure 1. Sampling distribution of $\hat{S}$

Figure 2. Approach of $\hat{S}$ to nN

We would therefore expect $\text{var}(\hat{S})$ to tend to $\text{var}(n_N)$, and not to zero, as the asymptotic formula which does not take account of the variation in n_N , does. From (A.4), the variance of n_N tends to $\frac{R_{\gamma+1}(\omega) - R_{\gamma}(\omega)}{\xi R_{\gamma}^2(\omega)}$ as $N \rightarrow \infty$. For our chosen set of parameter values, $R_{-\frac{1}{2}}(\cdot) = 1$, $R_{\frac{1}{2}}(\omega) = 1 + \frac{1}{\omega}$, $\omega \simeq 1$, and $\text{var}(n_N) \rightarrow \xi^{-1} = 1000$ (cf. p. 13). $\text{SE}(n_N)$ and the observed $\text{SE}(\hat{S})$ are plotted in figure 3.

So while $E(\text{asvar}(\hat{S}|\gamma))$ is a reasonable approximation for moderate sample sizes, for large samples the variation in n_N which we have failed to take account of in the m.l. estimation becomes significant. This could also be a reason for the apparent small upward bias of $\hat{S}$ for large n_N .

Computation was performed using S-Plus 3.3 on a Sun Sparc 20.

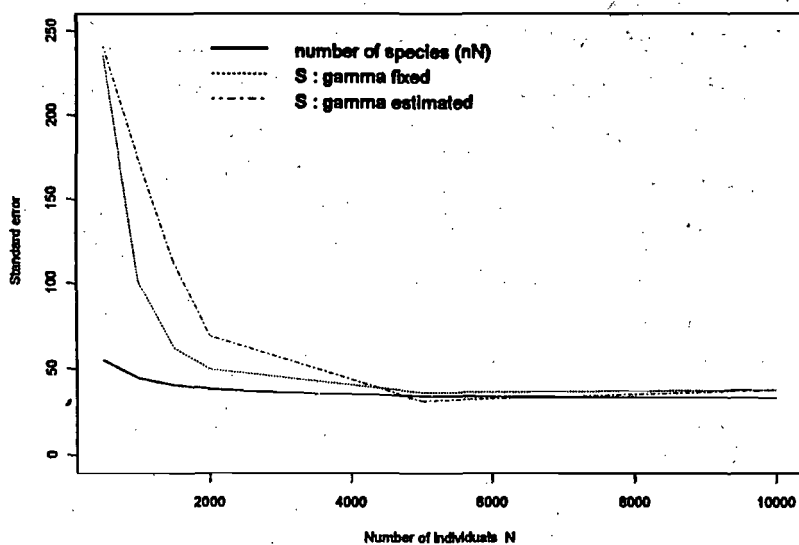


Figure 3. $SE(n_N)$ and simulated $SE(\hat{S})$

6. Conclusion

While many estimators for the total number of classes have been proposed (Burge & Fitzpatrick, 1993), those based on the GIGP are appealing because of the demonstrated flexibility of this distribution in many diverse applications (for example, Sichel, 1973, 1975, 1982; Burrell & Fenton, 1993; Willmot 1987). We have demonstrated the equivalence of two apparently different estimators proposed by Sichel (1986a, 1986b, 1992) and Ord and Whitmore (1986), given a technique for maximum likelihood estimation for the zero-truncated GIGP, and evaluated the asymptotic and small-sample behaviour of the estimator. While the estimator has been shown to be reasonably unbiased for large sample sizes, the asymptotic formula given for the variance underestimates the true variation as it fails to take account of the design in which Σx_i , rather than n , is fixed. Further work needs to be done in this area.

7. Appendix

A. DISTRIBUTION OF n_N

Consider the sum of independent random variables $Y = \sum_{i=1}^n X_i$, where n is fixed and known, $E(X_i) = \mu$ and $Var(X_i) = \sigma^2$.

Then $Y \sim N(n\mu, n\sigma^2)$ asymptotically, for large n .

$$\begin{aligned} f(y|n) &= \frac{1}{\sqrt{n}\sigma\sqrt{2\pi}} \exp \left\{ -\frac{1}{2} \left(\frac{y - n\mu}{\sqrt{n}\sigma} \right)^2 \right\} \\ &= \frac{1}{\sigma\sqrt{2\pi}} e^{\mu y / \sigma^2} n^{-\frac{1}{2}} \exp \left\{ -\frac{y\mu}{2\sigma^2} \cdot \left[\frac{y}{\mu} \cdot \frac{1}{n} + \frac{\mu}{y} \cdot n \right] \right\} \end{aligned} \quad (A.1)$$

Now we consider the distribution of the random variable N , where $y = \sum_{i=1}^N X_i$ is fixed and known. From (A.1),

$$\begin{aligned} g(n|y) &\propto f(y|n) \\ &= \mu \cdot \frac{\xi^{*-1/2}}{2K_{1/2}(\omega^*)} \cdot n^{-1/2} \cdot \exp\left[-\frac{\omega^*}{2} \left(\frac{\xi^*}{n} + \frac{n}{\xi^*}\right)\right] \end{aligned} \quad (\text{A.2})$$

where $\xi^* = y/\mu$ and $\omega^* = y\mu/\sigma^2$.

(A.2) is proportional to the probability function of a $\text{GIG}(\omega^*, \xi^*, \gamma = \frac{1}{2})$ random variate (or a reciprocal inverse Gaussian).

We use the notation

$$R_\nu(z) = \frac{K_{\nu+1}(z)}{K_\nu(z)} \quad (\text{as in Jorgensen, 1982}).$$

In the case of the X_i having the $\text{GIGP}(\omega, N\xi, \gamma)$ distribution,

$$\mu = N\xi R_\gamma(\omega) \text{ and } \sigma^2 = \mu\{1 + N\xi[R_{\gamma+1}(\omega) - R_\gamma(\omega)]\}.$$

So we have that

$$n_N \sim \text{GIG}(\omega^*, \xi^*, \frac{1}{2})$$

where

$$\xi^* = \frac{1}{N\xi R_\gamma(\omega)}$$

and

$$\omega^* = \frac{N}{1 + N\xi[R_{\gamma+1}(\omega) - R_\gamma(\omega)]}$$

Jorgensen (1982) gives the asymptotic distribution of $X \sim \text{GIG}(\omega, \xi, \gamma)$, as $\omega \rightarrow \infty$, as $\sqrt{\omega}(X/\xi - 1) \rightarrow N(0, 1)$.

ω^* increases with N to the limit $\{\xi[R_{\gamma+1}(\omega) - R_\gamma(\omega)]\}^{-1}$ which, with the typically very small values for ξ in these problems, satisfies the above condition.

Thus we have that $n_N \rightarrow N(\mu_{n_N}, \sigma_{n_N}^2)$, where

$$\begin{aligned}\mu_{n_N} &= \frac{1}{\xi R_\gamma(\omega)} \\ &= S\end{aligned}\quad (\text{Sichel 1986a}) \quad (\text{A.3})$$

and

$$\sigma_{n_N}^2 = \frac{1 + N\xi[R_{\gamma+1}(\omega) - R_\gamma(\omega)]}{N\{\xi R_\gamma(\omega)\}^2} \quad (\text{A.4})$$

B. DERIVATIONS

From (1) and (2) the probability function of the zero-truncated GIGP is

$$\begin{aligned}P^*(X_i = x) &= \left(\frac{1}{1 - p_0}\right) \frac{g_{x+\gamma}(\alpha)}{g_\gamma(\omega)} \frac{(\xi\omega)^x}{x!} \\ &= \frac{g_{x+\gamma}(\alpha)}{[g_\gamma(\omega) - g_\gamma(\alpha)]} \frac{(\xi\omega)^x}{x!} \quad x = 1, 2, \dots\end{aligned}$$

giving the log likelihood

$$\begin{aligned}L &= -n_N \log[g_\gamma(\omega) - g_\gamma(\alpha)] + n_N \bar{x} \log(\xi\omega) \\ &\quad + \sum \log g_{x_i+\gamma}(\alpha) - \sum \log x_i!\end{aligned}$$

We need the following derivatives:

$$\frac{\partial K_\nu(z)}{\partial z} = \frac{K_{\nu+1}(z) + K_{\nu-1}(z)}{-2}$$

from which

$$g'_\nu(z) = -zg_{\nu+1}(z)$$

and

$$\frac{\partial \log g_\nu(z)}{\partial z} = -R_\nu(z)$$

$$\text{Also, } R'_\nu(z) = R_\nu^2(z) - \frac{2\nu+1}{z} R_\nu(z) - 1$$

For simplification we define

$$h_\gamma(\alpha, \xi) = \frac{g_\gamma(\alpha) R_\gamma(\omega)}{\sqrt{\xi^2 + \alpha^2}}$$

$$\text{and } d_\gamma(\alpha, \xi) = g_\gamma(\omega) - g_\gamma(\alpha)$$

and we have

$$\frac{\partial h}{\partial \alpha} = \frac{\alpha[g_\gamma(\alpha) R'_\gamma(\omega) - \sqrt{\xi^2 + \alpha^2} g_{\gamma+1}(\alpha) R_\gamma(\omega) - h_\gamma(\alpha, \xi)]}{\xi^2 + \alpha^2}$$

$$\frac{\partial h}{\partial \xi} = \frac{-\omega R'_\gamma(\omega) g_\gamma(\alpha) - \xi h_\gamma(\alpha, \xi)}{\xi^2 + \alpha^2}$$

$$\frac{\partial d}{\partial \alpha} = -\omega g_{\gamma+1}(\omega) \frac{\alpha}{\sqrt{\xi^2 + \alpha^2}} + \alpha g_{\gamma+1}(\alpha)$$

$$\frac{\partial d}{\partial \xi} = \frac{\omega^2 g_{\gamma+1}(\omega)}{\sqrt{\xi^2 + \alpha^2}}$$

Straightforward algebra yields the following derivatives:

$$\frac{\partial L}{\partial \alpha} = \frac{-n_N \frac{\partial d}{\partial \alpha}}{d_\gamma(\alpha, \xi)} + \frac{n_N \bar{x} \alpha}{\omega \sqrt{\xi^2 + \alpha^2}} - \sum_{i=1}^{n_N} R_{x_i + \gamma}(\alpha) \quad (\text{B.1})$$

$$\frac{\partial L}{\partial \xi} = \frac{n_N \omega}{\sqrt{\xi^2 + \alpha^2}} \left\{ \bar{x} - \frac{\omega g_{\gamma+1}(\omega)}{d_\gamma(\alpha, \xi)} \right\} \quad (\text{B.2})$$

The second derivatives get rather messy:

$$\begin{aligned}
 \frac{\partial^2 L}{\partial \alpha^2} = & \frac{n_N \bar{x} [\omega(\xi^2 + \alpha^2) + \alpha^2 \xi - 2\alpha^2 \sqrt{\xi^2 + \alpha^2}]}{\omega^2 (\xi^2 + \alpha^2)^{1.5}} \\
 & + n_N R_\gamma(\omega) \frac{[\xi^2 \omega - \alpha^2 (2\gamma + 1) \sqrt{\xi^2 + \alpha^2}]}{\omega (\xi^2 + \alpha^2)^{1.5}} \\
 & + \frac{n_N \alpha^2}{\xi^2 + \alpha^2} (R_\gamma^2(\omega) - 1) + \frac{2}{\alpha} \frac{\sum_{i=1}^{n_N} x_i R_{x_i + \gamma}(\alpha)}{\sum_{i=1}^{n_N} R_{x_i + \gamma}(\alpha)} + \frac{2\gamma + 1}{\alpha} \frac{\sum_{i=1}^{n_N} R_{x_i + \gamma}(\alpha)}{\sum_{i=1}^{n_N} R_{x_i + \gamma}(\alpha)} \\
 & - \frac{\sum_{i=1}^{n_N} R_{x_i + \gamma}^2(\alpha)}{\sum_{i=1}^{n_N} R_{x_i + \gamma}(\alpha)} + n_N \\
 & + \frac{n_N}{d_\gamma(\alpha, \xi)^2} \{ d_\gamma(\alpha, \xi) [h_\alpha(\alpha, \xi) - g_{\gamma+1}(\alpha) + \alpha \left(\frac{\partial h}{\partial \alpha} + \alpha g_{\gamma+2}(\alpha) \right)] \quad (B.3) \\
 & - \alpha [h_\gamma(\alpha, \xi) - g_{\gamma+1}(\alpha)] \frac{\partial d}{\partial \alpha} \}
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial^2 L}{\partial \xi^2} = & n_N \left\{ \bar{x} \left[\frac{-1}{\xi^2} + \frac{\xi}{(\xi^2 + \alpha^2)^{1.5}} \right] + \frac{\omega^2}{\xi + \alpha^2} [R_\alpha^2(\omega) - 1] \right. \\
 & \left. - \frac{R_\alpha(\omega) [(2\gamma + 1) \omega \sqrt{\xi^2 + \alpha^2} - \alpha^2]}{(\xi^2 + \alpha^2)^{1.5}} \right. \\
 & \left. + \frac{[d_\gamma(\alpha, \xi) \omega \left[\frac{\partial h}{\partial \xi} + \frac{h_\gamma(\alpha, \xi)}{\sqrt{\xi^2 + \alpha^2}} \right] + \frac{\omega^3 h_\alpha(\alpha, \xi) g_{\gamma+1}(\omega)}{\sqrt{\xi^2 + \alpha^2}}]}{d_\gamma(\alpha, \xi)^2} \right\} \quad (B.4)
 \end{aligned}$$

and finally

$$\begin{aligned} \frac{\partial^2 L}{\gamma_\alpha \partial \xi} = n_N \left\{ \frac{\bar{x}}{(\xi^2 + \alpha^2)^{1.5}} - \frac{\alpha \omega}{\xi^2 + \alpha^2} [R_\alpha^2(\omega) - 1] \right. \\ \left. + \frac{\alpha[(2\gamma + 1)\sqrt{\xi^2 + \alpha^2} - \xi]}{(\xi^2 + \alpha^2)^{1.5}} R_\gamma(\omega) \right. \\ \left. + \frac{d_\gamma(\alpha, \xi) \left(\frac{-\alpha h_\gamma(\alpha, \xi)}{\sqrt{\xi^2 + \alpha^2}} - \omega \frac{\partial h}{\partial \alpha} \right) + \omega h_\gamma(\alpha, \xi) \frac{\partial d}{\partial \alpha}}{d_\gamma(\alpha, \xi)^2} \right\} \quad (\text{B.5}) \end{aligned}$$

In order to get the expectations of the second derivatives above we need the following:

$$\begin{aligned} E(X) &= \xi \omega \frac{g_{\gamma+1}(\omega)}{d_\gamma(\alpha, \xi)} \\ E(R_{X+\gamma}(\alpha)) &= \alpha \frac{d_{\gamma+1}(\alpha, \xi)}{d_\gamma(\alpha, \xi)} \\ E(X R_{X+\gamma}(\alpha)) &= \alpha \xi \omega \frac{g_{\gamma+2}(\omega)}{d_\gamma(\alpha, \xi)} \end{aligned}$$

$E(R_{X+\gamma}^2(\alpha))$ does not simplify to a closed form, so has to be computed numerically.

We then have the elements of the Fisher information matrix $\mathcal{I}$

$$\begin{aligned} \frac{1}{n_N} E \left[\frac{\partial^2 L}{\partial \alpha^2} \right] &= \frac{\alpha^2}{\xi^2 + \alpha^2} [R_\gamma^2(\omega) - 1] + 1 - E(R_{X+\gamma}^2(\alpha)) \\ &+ \frac{R_\gamma(\omega) [\xi^2 \omega - \alpha^2 (2\gamma + 1) \sqrt{\xi^2 + \alpha^2}]}{\omega (\xi^2 + \alpha^2)^{1.5}} \end{aligned}$$

$$\begin{aligned}
& + \frac{2\xi\omega g_{\gamma+2}(\omega)}{d_\gamma(\alpha, \xi)} + \frac{(2\gamma+1)d_{\gamma+1}(\alpha, \xi)}{d_\gamma(\alpha, \xi)} \\
& + \frac{\xi\omega g_{\gamma+1}(\omega)}{d_\gamma(\alpha, \xi)} \frac{\omega(\xi^2 + \alpha^2) + \alpha^2\xi - 2\alpha^2\sqrt{\xi^2 + \alpha^2}}{\omega^2(\xi^2 + \alpha^2)^{1.5}} \\
& + \frac{d_\gamma(\alpha, \xi) \frac{\partial a_\gamma(\alpha, \xi)}{\partial \alpha} - a_\gamma(\alpha, \xi) \frac{\partial d_\gamma(\alpha, \xi)}{\partial \alpha}}{[d_\gamma(\alpha, \xi)]^2} \\
& = -\frac{1}{n_N} \mathcal{I}_{11}
\end{aligned}$$

$$\text{where } a_\gamma(\alpha, \xi) = \alpha(h_\gamma(\alpha, \xi) - g_{\gamma+1}(\alpha))$$

$$\text{and } \frac{\partial a_\gamma(\alpha, \xi)}{\partial \alpha} = h_\gamma(\alpha, \xi) - g_{\gamma+1}(\alpha) + \alpha \left(\frac{\partial h}{\partial \alpha} + \alpha g_{\gamma+2}(\alpha) \right)$$

$$\begin{aligned}
\frac{1}{n_N} E \left[\frac{\partial^2 L}{\partial \xi^2} \right] & = E(X) \left[\frac{-1}{\xi^2} + \frac{\xi}{(\xi^2 + \alpha^2)^{1.5}} \right] + \left(\frac{w^2}{\xi^2 + \alpha^2} \right) (R_\gamma^2(\omega) - 1) \\
& - R_\gamma(w) ((2\gamma+1)\omega\sqrt{\xi^2 + \alpha^2} - \alpha^2) / (\xi^2 + \alpha^2)^{1.5} \\
& + \left[d_\gamma(\alpha, \xi) \cdot \frac{\partial c}{\partial \xi} - c_\gamma(\alpha, \xi) \frac{\partial d}{\partial \xi} \right] / [d_\gamma(\alpha, \xi)]^2 \\
& = -\frac{1}{n_N} \mathcal{I}_{22}
\end{aligned}$$

$$\text{where } c_\gamma(\alpha, \xi) = -\omega h_\gamma(\alpha, \xi)$$

$$\text{and } \frac{\partial c}{\partial \xi} = \omega \left[-\frac{\partial h}{\partial \xi} + h_\gamma(\alpha, \xi) / \sqrt{\xi^2 + \alpha^2} \right]$$

$$\frac{\partial c}{\partial \alpha} = \frac{-\alpha h_\gamma(\alpha, \xi)}{\sqrt{\xi^2 + \alpha^2}} - \omega \frac{\partial h}{\partial \alpha}$$

and the asymptotic variances and covariance are given by the usual formulae

$$asvar(\hat{\alpha}|\gamma, n_N) = \frac{I_{22}}{\det(I)} \quad (B.6)$$

$$asvar(\hat{\xi}|\gamma, n_N) = \frac{I_{11}}{\det(I)} \quad (B.7)$$

$$ascov(\hat{\alpha}, \hat{\xi}|\gamma, n_N) = \frac{-I_{12}}{\det(I)} \quad (B.8)$$

$$\text{where } \det(I) = I_{11}I_{22} - I_{12}^2$$

8. References

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